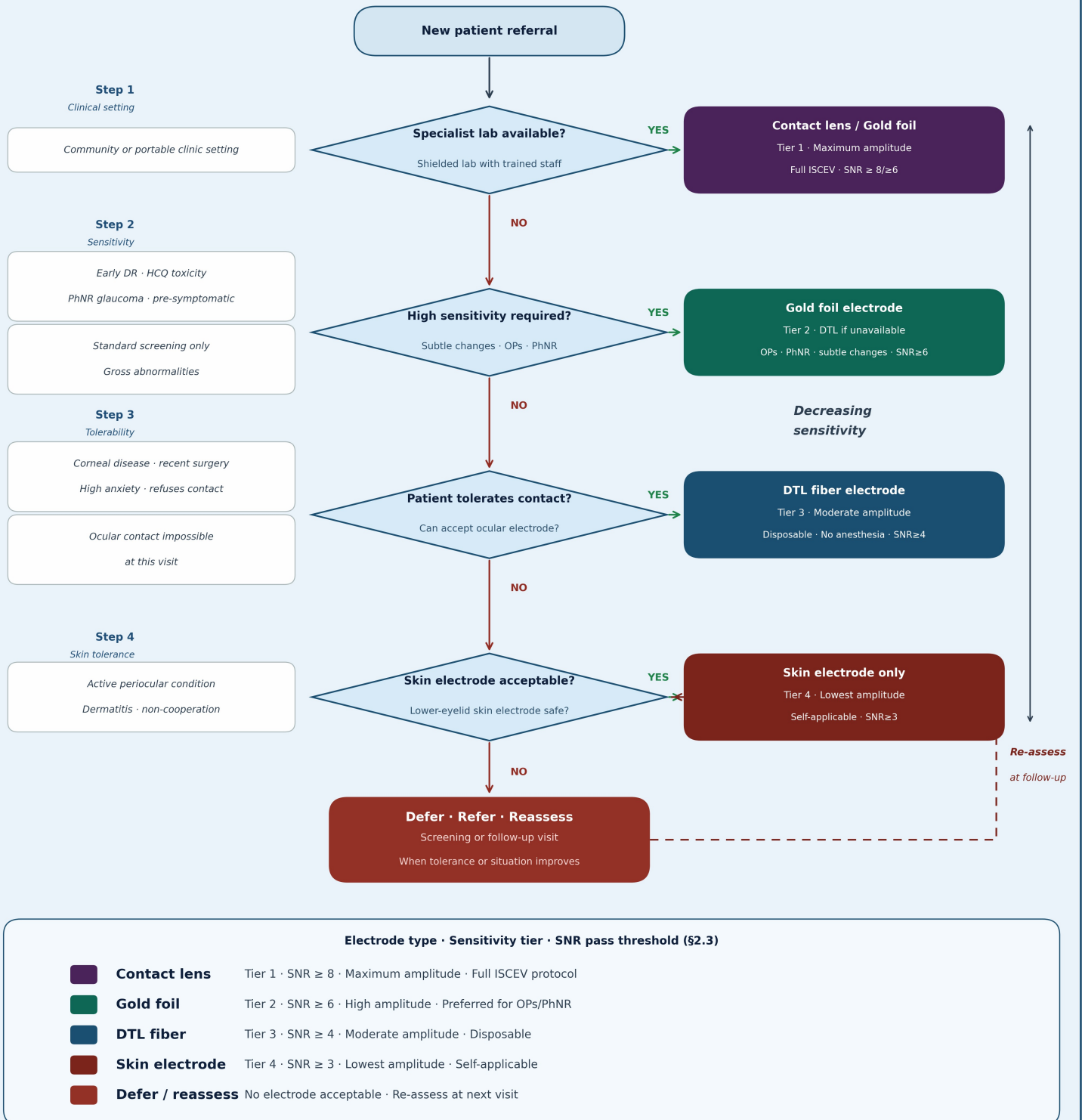


Figure 2.1 — Clinical decision algorithm for electrode selection

Three sequential decision nodes narrow the choice to the single best electrode



Dashed return path indicates deferred decision pending re-assessment at follow-up.

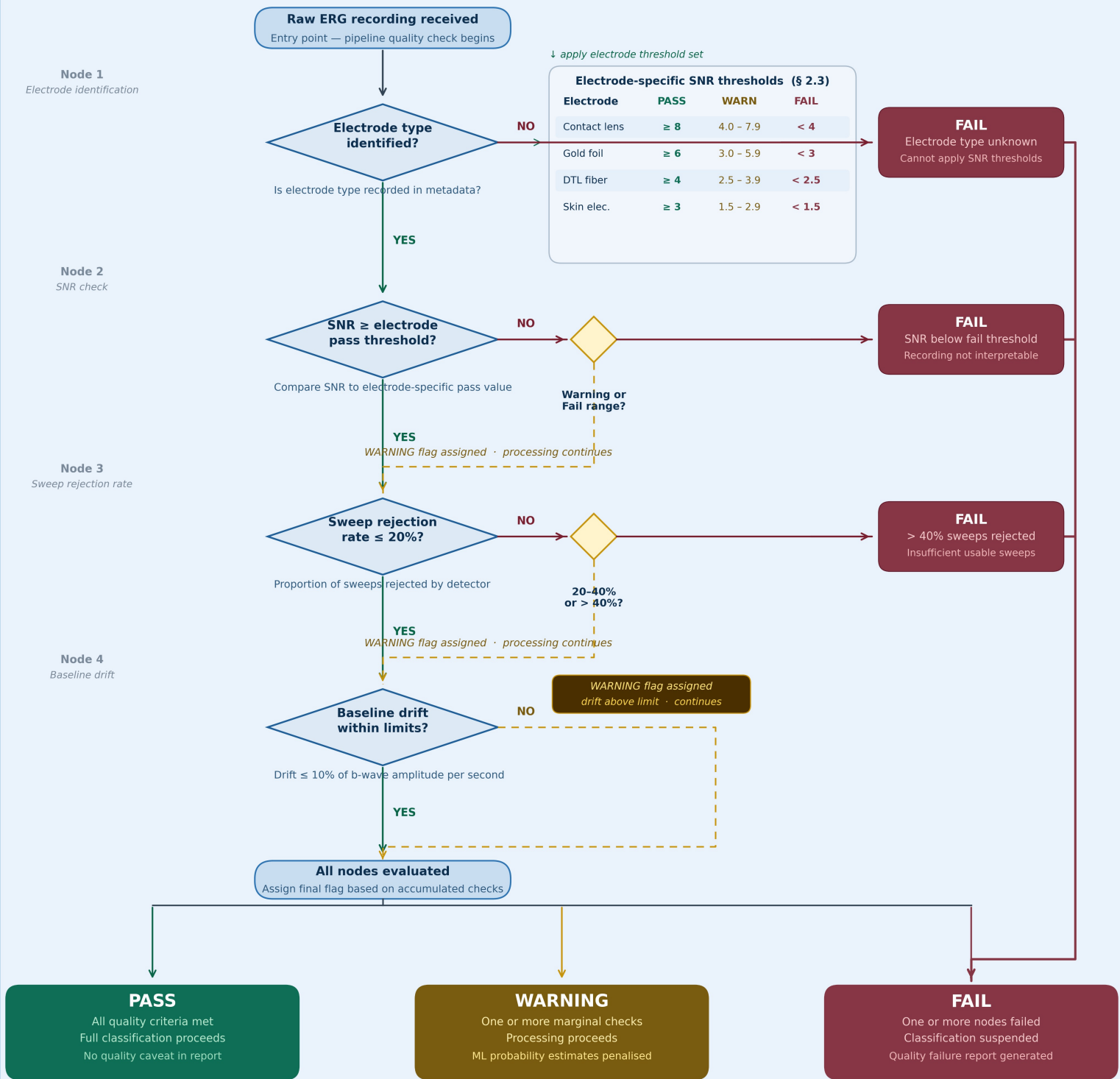
Abbreviations: ISCEV = International Society for Clinical Electrophysiology of Vision;

SNR = signal-to-noise ratio; DR = diabetic retinopathy; HCQ = hydroxychloroquine;

OP = oscillatory potential; PhNR = photopic negative response.

Figure 2.2 — Signal quality flag decision tree

Applied sequentially to every incoming ERG recording · Exit flags: PASS, WARNING, or FAIL



Pipeline action for each flag (§ 2.5)

- PASS** Full preprocessing and classification proceed. No quality caveat in clinical report.
- WARNING** Processing proceeds. ML probability estimates calibration-adjusted downward. Quality caveat appended to report.
- FAIL** Classification suspended. Quality failure report generated. Corrective actions recommended. Event logged to rejection register.

Decision nodes are applied sequentially to every incoming recording. FAIL at any node suspends processing immediately. WARNING is recorded as metadata and processing continues.

Abbreviations: SNR = signal-to-noise ratio; RMS = root-mean-square; ML = machine learning; ERG = electroretinogram.

SNR thresholds — Contact lens: Pass ≥ 8 , Warn 4–7.9, Fail < 4 | Gold foil: Pass ≥ 6 , Warn 3–5.9, Fail < 3 | DTL fiber: Pass ≥ 4 , Warn 2.5–3.9, Fail < 2.5

Skin electrode: Pass ≥ 3 , Warn 1.5–2.9, Fail < 1.5 .